TTK4190 Assignment 2 Part 2
Heading Autopilot & Speed Control

Objective

In this assignment, you will design a heading autopilot and a speed controller to steer the marine craft model from Assignment 2 Part 1. The autopilot will be tested by adding environmental disturbances in the form of current and wind. In problem 3 of this assignment, you are going to implement a propulsion system and a speed controller for the ship model. To solve this task, you will have to simulate additional states; do this by extending the state vector as needed.

For this assignment, you will use the new template files `ship.m` and `main.m` provided to you on Canvas do not reuse the `ship.m` and `main.m` files from the previous assignment. The template files contain comments that can assist you in structuring your code. While you're not obligated to follow the comments precisely, keeping a similar structure will make it easier for us to help you with debugging. The code you produce in this assignment will be further developed in Part 3. Therefore, make sure that your code is easy to read, update, and debug.

Delivery Details

The tasks may be completed individually or in groups. The hand-in should be a PDF report and the code (.m files) for each sub-problem. Upload the .m files directly, not in a zip folder.

Problem 1: Environmental Disturbances

(a) Download the newest versions of `main.m` and `ship.m` from Canvas and modify the code to include 2-D irrotational ocean current in surge and sway. Let the current speed be $V_c = 1$ m/s and assume the current is moving towards the north-east direction, i.e., $\beta_{V_c} = 45°$.
Hint: Sec. 10.3.2 in (Fossen, 2021).

$$u_c = V_c \cos(\beta_{V_c} - \psi)$$
$$v_c = V_c \sin(\beta_{V_c} - \psi)$$

(b) Simulate the ship model with and without current and explain the observed deviation between the sideslip and crab angles. Include a figure comparing the sideslip and the crab angle in your report.

(c) Update the code to include wind moments Y_{wind} and N_{wind} in sway and yaw, respectively, occurring after 200 seconds. Let the wind speed be $V_w = 10$ m/s and assume the wind is moving towards the southeast direction, i.e., $\beta_{V_w} = 135°$. Use that $\rho_a = 1.247$ kg/m^3, $c_y = 0.95$, $c_n = 0.15$, and $A_{L_w} = 10 L_{oa}$, where L_{oa} is the length of the ship.

Hint: Eqs. (10.21–10.23) in (Fossen, 2021), wind coefficient approximation for symmetrical ships:

$$C_X(\gamma_{\text{w}}) \approx -c_x \cos(\gamma_{\text{w}})$$
$$C_Y(\gamma_{\text{w}}) \approx c_y \sin(\gamma_{\text{w}})$$
$$C_N(\gamma_{\text{w}}) \approx c_n \sin(2\gamma_{\text{w}})$$

Problem 2: Heading Autopilot

(a) Linearize the nonlinear Coriolis forces $\mathbf{C}_{\text{RB}}(\boldsymbol{\nu})\boldsymbol{\nu}$ and $\mathbf{C}_{\mathbf{A}}(\boldsymbol{\nu})\boldsymbol{\nu}$ derived in the previous assignment about $v = r = 0$ and $u = u_d$. Write down the final expressions, denoting them as $\mathbf{C}^*_{\text{RB}}$ and $\mathbf{C}^*_{\text{A}}$.

(b) Using the linear sway-yaw maneuvering model

$$\mathbf{M}\dot{\boldsymbol{\nu}}_r + \mathbf{N}\boldsymbol{\nu}_r = \mathbf{b}\delta$$

from eq. (6.138) in (Fossen, 2021), write down the transfer function from δ to r numerically. Note that $\mathbf{M} := \mathbf{M}_{\text{RB}} + \mathbf{M}_{\text{A}}$, $\mathbf{N} := \mathbf{C}^*_{\text{RB}} + \mathbf{C}^*_{\text{A}} + \mathbf{D}$, and $\mathbf{b} = \mathbf{b}^* = 2u_d[-Y_\delta,\ -N_\delta]^\top$, where Y_δ and N_δ have already been implemented in `ship.m`.

Hint: The Matlab function `ss2tf(.)` can be used to transform a state-space model to a transfer function. Furthermore, $\mathbf{M}^{-1}$ is available in `ship.m`, you just need to extract sway and yaw dynamics.

(c) Based on the transfer function in (b), write down numerical values for T and K corresponding to a first-order Nomoto model.

Hint: Eqs. (7.20), (7.24), and (7.25) in (Fossen, 2021).

$$\frac{r}{\delta}(s) = \frac{K(T_3 s + 1)}{(T_1 s + 1)(T_2 s + 1)}$$

The Laplace transform can be approximated as

$$\frac{r}{\delta}(s) = \frac{K}{s(Ts + 1)}$$
$$T := T_1 + T_2 - T_3$$

(d) Design a PID controller using pole placement (Alg. 15.1, Fossen, 2021) with $\omega_b = 0.06$ rad/s and $\zeta = 1$. Using plots show that the controller can compensate for environmental disturbances. Use a 3rd-order reference model with $\omega_{\text{ref}} = 0.03$ rad/s to generate the desired yaw angle and yaw rate. *Hint:* Eq. (15.143) in (Fossen, 2021)

$$\frac{\psi_d}{\psi_{\text{ref}}}(s) = \frac{\omega_n^3}{(s + \omega_n)(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

(e) Demonstrate the controller performance for a $10°$ heading setpoint followed by a $-20°$ heading setpoint. Attach a plot showing the desired and actual heading. Would you claim that integrator windup is a problem in your simulations? If so, implement an anti-windup strategy.

Problem 3: Propeller Revolution and Speed Control

The block diagram shown in Figure 1 represents the propulsion system that we are going to implement. The material relevant to the following problems can be found in Chapter 9 (Fossen, 2021).

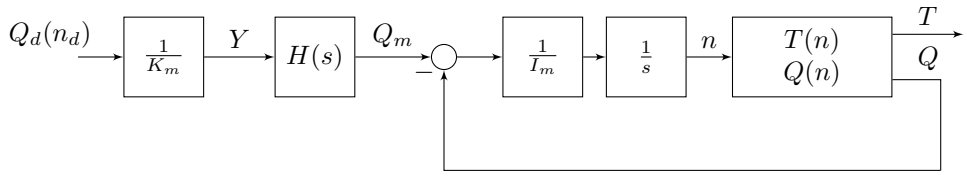


Figure 1: A block diagram showing the dynamics of the prime mover system in the marine craft.

(a) Let $J_a = 0$ and use the Matlab function `[KT,KQ] = wageningen(.)` to compute the propeller thrust and torque coefficients K_T and K_Q, respectively, for a 4-blade propeller with a pitch/diameter ratio of 1.5 and a blade area ratio of 0.65. Update your code to use the new values for K_T and K_Q.

Note: Setting the advance number $J_a = 0$ corresponds to the ship being stationary while the propeller rotates. In reality, propeller performance depends on the ship's forward speed relative to the water. This is a useful simplification for this exercise, but keep in mind that the true thrust calculation is more complicated when the ship is operating at different speeds.

(b) Given a desired motor shaft speed n_d, how can we find the equivalent motor torque needed to generate that speed? Find this relation and describe the desired propeller torque Q_d as a function of the desired shaft speed n_d. $Q_d(n_d) = ...$ This relation is needed to complete the next task.
Hint: Eq. (9.8) in (Fossen, 2021)

$$Q = \rho D^5 K_Q(J_a)|n|n$$

(c) In `ship.m`, implement the propulsion system shown in Figure 1, where T and Q are given by eqs. (9.7) and (9.8) in (Fossen, 2021). The formula for the shaft speed dynamics $\dot{n}$ should be updated accordingly by using $I_m = 100\ 000$ kg m^2, $K_m = 0.6$ s^{-1}, $T_m = 10$ s, and $\tau = 0$ s. You might need to make changes in `main.m` to include additional states.

Hint: Eqs. (9.33)in (Fossen, 2021)

$$I_m \dot{n} = Q_m - Q - Q_f$$

where Q_m is produced torque developed by the main engine and Q_f is the friction torque.
Eq (9.35) in (Fossen, 2021)

$$\frac{Q_m}{Y} = H(s) = \frac{K_m}{sT_m + 1} e^{-\tau s} \tag{1}$$

(d) Under which assumptions will the ship cruise speed satisfy $U = \frac{(t-1)T}{X_u}$?
Hint: Eq. (6.136) in (Fossen, 2021), linear forward speed model

$$(m - X_{\dot{u}})\dot{u} - X_u u = -X_{\delta\delta}\delta^2 + (1-t)T$$

(e) Change the desired cruise speed to $u_d = U_{\text{ref}} = 9$ m/s and use the equation from (d) to compute the desired thrust T_d and propeller revolution n_d. Include these expressions in the code and simulate the ship. Do you achieve the desired speed if the heading angle is constant? Explain why/why not.

(f) **(Optional)** Propose modifications to (e) to ensure that the marine craft reaches the desired cruise speed in steady-state. Attach a plot showing that the desired and actual cruise speeds coincide and explain the changes you have made.